

KANSK, Russian Federation

WMO: 295810

Lat: 56.2017N

Lon: 95.6483E

Elev: 203

StdP: 98.91

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.7	-37.5	-43.2	0.1	-40.5	-40.2	0.1	-37.4	11.4	-9.3	10.2	-10.8	0.6	120	0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	29.7	19.7	27.8	18.8	26.1	18.2	21.1	27.1	20.0	25.8	19.1	24.5	2.3	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	14.1	24.1	18.0	13.3	22.8	17.0	12.4	22.0	62.3	27.1	58.6	25.9	55.3	24.5	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.1	6.9	-41.6	32.8	3.9	1.8	-44.4	34.1	-46.7	35.2	-48.9	36.2	-51.7	37.5		
(5)			-40.8	23.0	4.1	1.2	-43.8	23.9	-46.2	24.6	-48.5	25.2	-51.5	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.2	-20.8	-17.0	-7.6	2.5	9.6	16.9	19.2	16.2	8.8	1.1	-9.6	-17.7
	DBStd	15.29	10.00	8.69	7.29	4.89	4.99	3.99	2.90	3.56	4.20	5.30	8.52	9.81
	HDD10.0	4359	956	756	546	229	71	2	0	2	72	278	588	858
	HDD18.3	6716	1214	989	804	476	273	72	25	86	287	535	838	1117
	CDD10.0	787	0	0	0	3	59	211	284	193	36	2	0	0
	CDD18.3	102	0	0	0	0	3	30	51	18	1	0	0	0
	CDH23.3	1117	0	0	0	2	52	396	454	198	17	0	0	0
	CDH26.7	276	0	0	0	0	8	112	106	47	3	0	0	0
Wind	WSAvg	2.4	2.2	2.3	2.7	3.2	3.1	2.1	1.8	1.7	2.2	2.8	2.7	2.4
Precipitation	PrecAvg	312	10	10	7	13	28	37	62	53	34	23	16	15
	PrecMax	442	31	30	22	41	63	72	144	123	68	68	31	34
	PrecMin	218	1	1	0	1	1	12	9	8	4	3	3	1
	PrecStd	65	7	6	4	9	16	15	34	26	17	13	7	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.6	2.3	9.2	20.4	27.8	31.9	31.7	30.6	25.5	16.6	9.3	2.7
		MCWB	-0.4	-0.6	4.1	10.1	15.7	19.3	21.3	20.6	16.6	9.6	5.6	0.1
	2%	DB	-1.2	0.1	6.1	16.3	24.7	29.6	29.5	27.8	21.5	13.3	5.7	-0.6
		MCWB	-2.8	-2.0	2.4	8.6	14.4	18.7	20.4	19.3	14.7	8.3	2.9	-2.2
	5%	DB	-3.7	-1.9	4.1	13.6	22.0	27.6	27.6	25.5	18.6	10.7	3.2	-2.4
		MCWB	-5.0	-3.6	1.0	7.1	12.7	18.1	19.6	18.4	13.3	7.0	1.1	-3.7
	10%	DB	-6.3	-4.5	2.1	11.0	19.3	25.5	26.0	23.5	15.9	8.5	1.3	-4.4
		MCWB	-7.4	-5.8	-0.3	5.9	11.4	17.4	18.9	17.3	11.9	5.6	-0.3	-5.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.5	-0.4	4.7	11.2	17.2	21.3	23.3	22.0	17.7	10.7	5.8	-0.1
		MCDB	1.4	1.7	8.4	17.8	24.8	27.8	29.2	28.5	23.5	15.1	9.0	2.4
	2%	WB	-2.8	-1.9	2.7	9.3	15.2	20.1	21.5	20.3	15.4	8.9	3.1	-2.1
		MCDB	-1.3	-0.1	5.7	15.3	22.8	26.9	27.3	26.4	20.6	12.1	5.2	-0.5
	5%	WB	-5.0	-3.6	1.2	7.6	13.7	19.1	20.5	19.0	13.7	7.5	1.3	-3.6
		MCDB	-3.7	-1.9	3.7	13.1	20.6	25.6	25.9	24.2	17.5	10.6	3.1	-2.4
	10%	WB	-7.4	-5.8	-0.1	5.9	12.2	18.1	19.6	17.9	12.3	5.7	-0.2	-5.6
		MCDB	-6.3	-4.5	2.0	10.8	18.3	24.4	24.7	22.5	15.4	8.2	1.2	-4.5
Mean Daily Temperature Range	5% DB	MDBR	8.6	10.3	11.3	10.3	11.5	11.8	10.6	10.7	10.1	8.6	7.8	8.4
		MCDBR	8.3	10.0	11.9	15.6	16.4	16.3	13.6	14.7	16.0	13.1	8.3	8.5
		MCWBR	7.4	8.5	8.7	8.9	8.0	7.0	6.0	7.3	9.0	8.4	6.3	7.6
	5% WB	MCDBR	8.4	10.2	11.5	14.3	14.2	13.5	11.5	13.5	13.8	12.5	7.7	8.5
		MCWBR	7.5	8.8	8.6	8.8	7.5	6.5	6.0	7.2	8.0	8.3	6.2	7.6
Clear-Sky Solar Irradiance	taub	0.280	0.313	0.354	0.389	0.411	0.414	0.430	0.409	0.365	0.345	0.310	0.279	
	taud	2.014	2.001	1.974	2.141	2.193	2.273	2.250	2.274	2.352	2.253	2.077	1.970	
	Ebn at Noon	577	713	780	822	827	829	805	797	784	685	533	459	
	Edh at Noon	65	99	132	131	132	124	125	114	92	78	62	53	
All-Sky Solar Radiation	RadAvg	0.79	1.79	3.20	4.38	5.03	5.95	5.45	4.40	2.78	1.50	0.75	0.48	
	RadStd	0.08	0.10	0.20	0.25	0.48	0.49	0.38	0.35	0.33	0.12	0.10	0.06	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page